

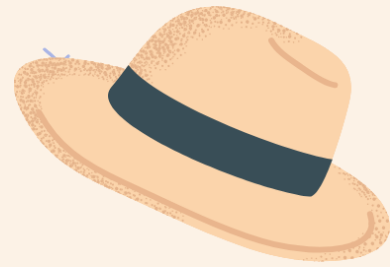
Hackathon 5 · SDG 8.9

Does building hotel rooms pay off?



Tourism investment and revenue

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AI for Good · Term 1, Week 5



Why it matters



3.7%

tourism's median share of GDP (2019, 136 countries)

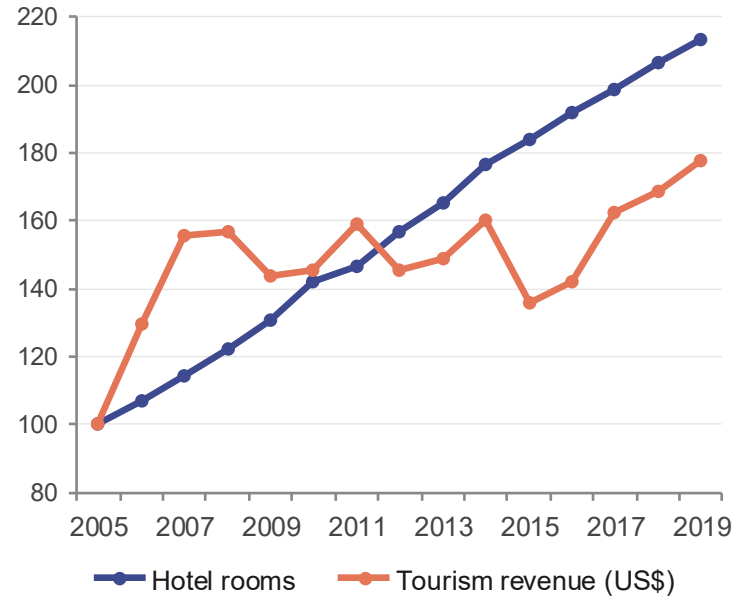
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countries where it's more than 10% of GDP

Governments back new hotels with permits, tax breaks and loans. But more rooms don't guarantee more visitors. If they don't come, you get empty hotels, debt and lost jobs.

Source: UN Tourism, SDG indicator 8.9.1

Morocco: rooms kept growing, revenue didn't
(2005 = 100)



Our user and question



User

Morocco's Ministry of Tourism, deciding whether to back more hotel building.



Question

A country added a lot of hotel rooms in the last 3 years. Will its tourism revenue grow faster than the world in the next 2?



Answer

Yes or no, with a probability. Used as a first screen, not the final decision.



The dataset

2,577

country-years

168

countries

1998–2017

years used



UN Tourism: hotel rooms, visitor spending, arrivals

World Bank: GDP, inflation, exchange rate, region, income group

One row = one country in one year

Left out: COVID years 2020–2022 and changes in how rooms are counted

Hotel rooms = our stand-in for investment

There's no free dataset with tourism investment money for all countries.

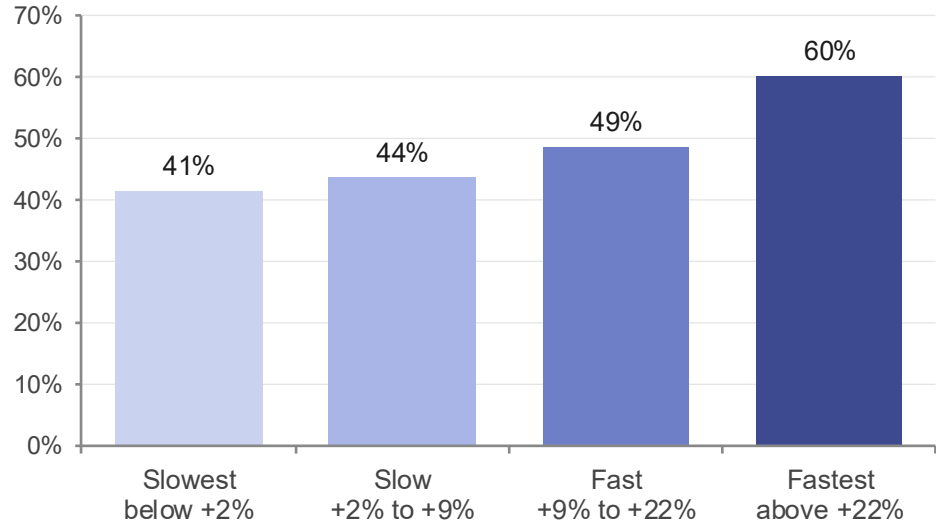
The idea in one chart

Fastest builders beat the world 60% of the time

The slowest group: 41%. There's a pattern, but it's weak. Even in the fastest group, 4 in 10 lag.



Share that beat the world, by room growth over 3 years
(training data)



The choice we'd defend



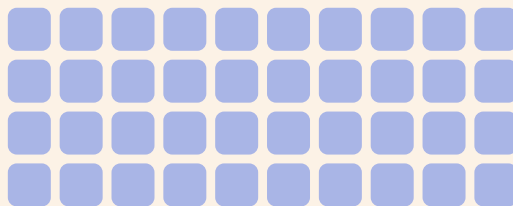
› Split by country, not by row

Morocco 2014 and 2015 share most of their data. A random split puts near-copies in train and test. So all years of a country go to one side.

› F1 as the main metric

A wrong yes wastes public money. A wrong no means jobs that never exist. F1 is only high if both are low.

135 countries

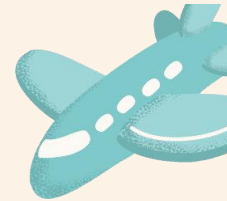


train + tuning (5 folds, by country)

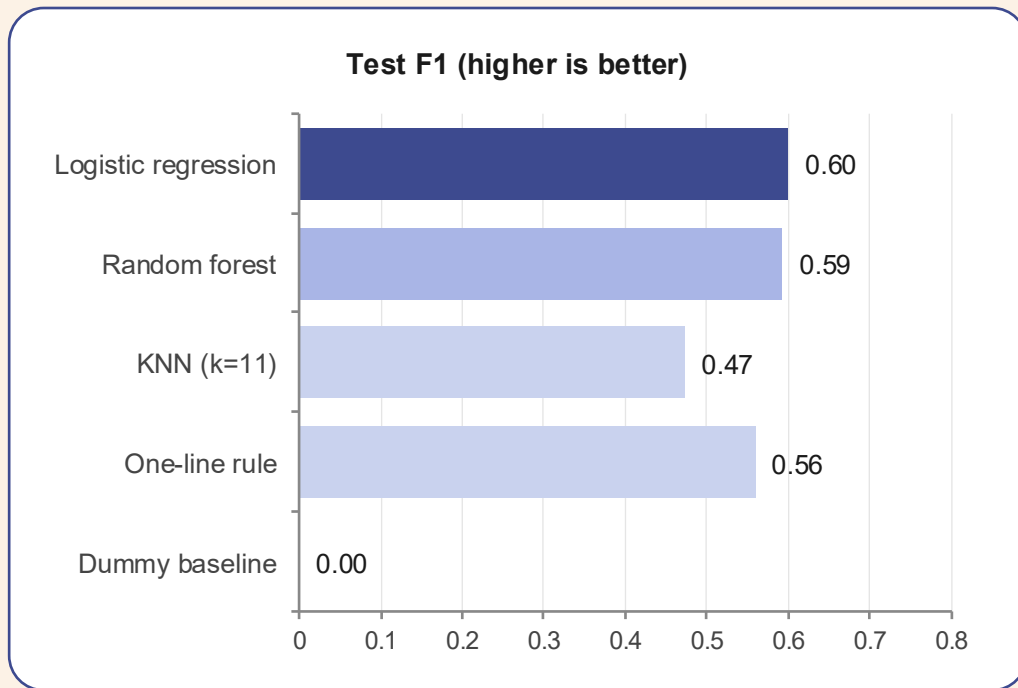
33 countries



test, used once, never seen in training



Results on unseen countries



Logistic regression wins: 0.60

Only a bit better than the one-line rule (0.56)

Random forest 0.59, but it memorises: train F1 0.91

KNN 0.47, worse than the rule

Our Choice

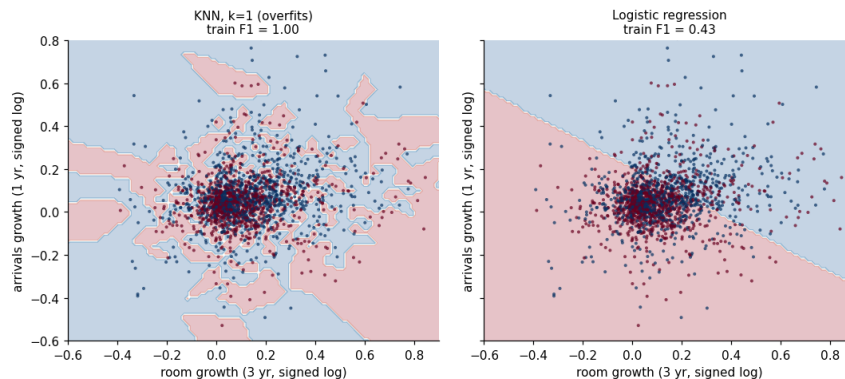
Logistic regression, test set

	Predicted: lags	Predicted: beats
Actually lags	141 right no	110 wrong yes
Actually beats	107 wrong no	163 right yes

Precision 0.60 · Recall 0.60

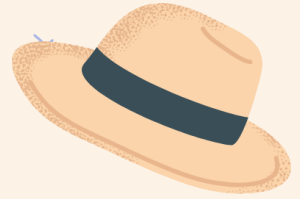


Each dot is a training country-year on two features. Blue = beat the world, red = didn't.

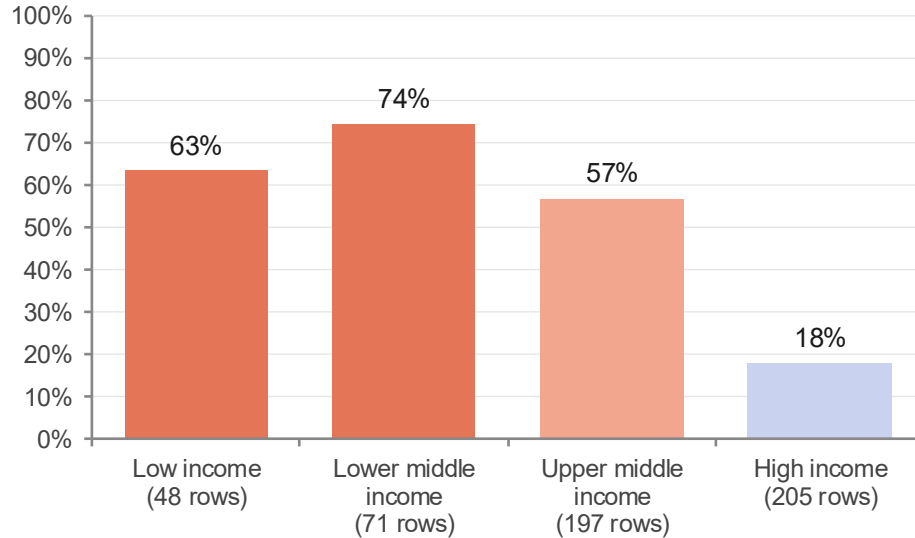


The dots overlap almost everywhere. KNN with $k=1$ draws islands around single dots (overfitting). Logistic regression draws one line. No model can split dots this mixed, so about 0.6 is close to what this data allows.

Riskier for poorer countries



Wrong yes rate: countries that lagged but got a yes



- **3 in 4** lower-middle income countries that lagged still got a yes. For rich countries: 1 in 5
- **Most rows are rich countries**; only 6% are low-income
- **What we did:** first screen only. For poorer countries a yes means "look closer"

Test with a made-up country

Lower-middle income, Middle East & North Africa

Hotel rooms +25% in 3 years

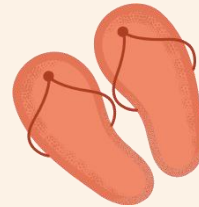
Arrivals +8%, revenue +6% last year

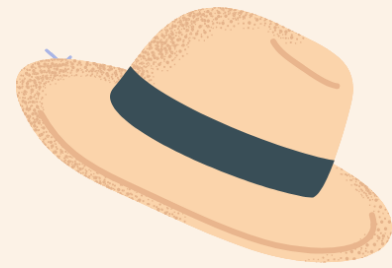
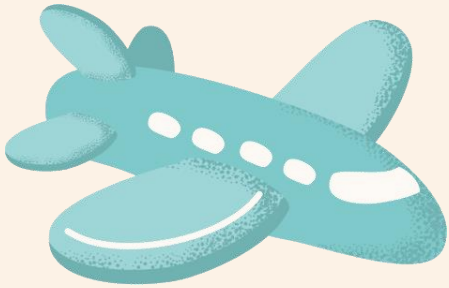
Currency 3% weaker, inflation 5%

52% chance to beat the world

Our advice

Use logistic regression as a first screen, never as the final decision. More rooms only move the chance a few points. Region and the economy matter at least as much.





Thank you

